

Period Finding using QFT

Introduction

An essential part of Shor's factorization algorithm is finding the **period** P of a function $f(x)$ such that:

$$f(x + P) = f(x)$$

This periodicity is extracted efficiently using the Quantum Fourier Transform (QFT).

Step 1: Initial State

We consider a system of **3 qubits** in the first register and **3 qubits** in the second register.

Initial state:

$$|000\rangle \otimes |000\rangle$$

Step 2: Apply Hadamard on First Register

Applying Hadamard transform on each of the 3 qubits:

$$|0\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

For 3 qubits:

$$\begin{aligned} |000\rangle &\rightarrow \left(\frac{1}{\sqrt{2}}\right)^3 \sum_{x=0}^7 |x\rangle \\ &= \frac{1}{\sqrt{8}}(|000\rangle + |001\rangle + \cdots + |111\rangle) \end{aligned}$$

Thus total state becomes:

$$\frac{1}{\sqrt{8}} \sum_{x=0}^7 |x\rangle \otimes |000\rangle$$

Note: The factor $\frac{1}{\sqrt{8}}$ ensures normalization since there are 8 equally probable states.

Step 3: Apply Oracle Function

The oracle computes:

$$|x\rangle|0\rangle \rightarrow |x\rangle|f(x)\rangle$$

Thus:

$$|\psi\rangle = \frac{1}{\sqrt{8}} \sum_{x=0}^7 |x\rangle |f(x)\rangle$$

Step 4: Apply QFT on First Register

The QFT transformation is:

$$|x\rangle \rightarrow \frac{1}{\sqrt{8}} \sum_{j=0}^7 e^{\frac{2\pi i x j}{8}} |j\rangle$$

Apply this to the entire state:

$$\begin{aligned} |\psi'\rangle &= \frac{1}{\sqrt{8}} \sum_{x=0}^7 \left(\frac{1}{\sqrt{8}} \sum_{j=0}^7 e^{\frac{2\pi i x j}{8}} |j\rangle \right) |f(x)\rangle \\ &= \frac{1}{8} \sum_{x=0}^7 \sum_{j=0}^7 e^{\frac{2\pi i x j}{8}} |j\rangle |f(x)\rangle \end{aligned}$$

Step 5: Rearranging the Expression

Rewriting the double sum:

$$|\psi'\rangle = \frac{1}{8} \sum_{j=0}^7 |j\rangle \left[\sum_{x=0}^7 e^{\frac{2\pi i x j}{8}} |f(x)\rangle \right]$$

Step 6: Using Periodicity

Assume period $P = 2$, so:

$$f(0) = f(2) = f(4) = f(6) = a$$

$$f(1) = f(3) = f(5) = f(7) = b$$

Step 7: Evaluate for Different j Values

Case $j = 0$

$$\begin{aligned}\sum_{x=0}^7 e^{0} f(x) &= f(0) + f(1) + \cdots + f(7) \\ &= 4a + 4b\end{aligned}$$

Thus:

$$\frac{1}{8}|0\rangle(4a + 4b)$$

Case $j = 1$

$$\sum_{x=0}^7 e^{\frac{2\pi i x}{8}} f(x)$$

Grouping:

$$\begin{aligned}&a\left(1 + e^{\frac{4\pi i}{8}} + e^{\frac{8\pi i}{8}} + e^{\frac{12\pi i}{8}}\right) \\ &+ b\left(e^{\frac{2\pi i}{8}} + e^{\frac{6\pi i}{8}} + e^{\frac{10\pi i}{8}} + e^{\frac{14\pi i}{8}}\right)\end{aligned}$$

Using:

$$1 + i + (-1) + (-i) = 0$$

Thus coefficient becomes:

$$0$$

Similarly

For many values of j , coefficients cancel out due to destructive interference.

Step 8: Final Result

Only specific values of j survive (constructive interference).

These satisfy:

$$\frac{j}{8} \approx \frac{k}{P}$$

For $P = 2$:

$$j = 0, 4$$

Conclusion

- QFT converts periodic structure into frequency peaks
- Constructive interference reveals correct values
- Measurement of j gives information about P

Final result

QFT transforms hidden periodicity into measurable frequency information

Understanding the Purpose of QFT in Period Finding

What may initially seem confusing is actually the central idea behind the entire procedure. In the worked example, the period $P = 2$ is assumed only for illustration, which can make it appear as if the computation is circular. However, in the actual problem—such as in Shor’s algorithm—the period is **not known in advance**. The goal of the algorithm is precisely to determine this unknown period.

The process begins by creating a superposition of all possible input states and then applying a function that is known to be periodic, although its period is hidden. This function introduces structure into the quantum system by correlating different input values that produce the same output. When the Quantum Fourier Transform (QFT) is applied to this state, it transforms the hidden periodic structure in the input domain into a frequency-domain representation, where distinct peaks emerge.

These peaks occur at specific values that are mathematically related to the unknown period. By measuring the system and performing classical post-processing, one can extract the value of the period. Therefore, the example using $P = 2$ is not solving the problem itself, but rather demonstrating that if a function has a period, the QFT will reveal it automatically through quantum interference.